

AROOJ FATIMA

afatima@nust.edu.pk[LinkedIn](#)[GitHub](#)[Codeberg](#)[Website](#)

EDUCATION

National University of Sciences & Technology (NUST)

Bachelor of Science in Bioinformatics

CGPA: 3.41/4.00

Islamabad, Pakistan

Sep 2023–Jun 2027

Relevant Coursework: Discrete Mathematics, Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Computer Organization and Assembly Language, Parallel & Distributed Computing, Linear Algebra, Probability and Statistics, Software Engineering, Fundamental of Computer Programming, Object Oriented Programming, Database Systems

RESEARCH INTERESTS

Formal Verification, Interactive Theorem Proving, Systems Software Verification, C Program Verification, Programming Languages and Formal Semantics, and Verified Compilation.

RESEARCH & PROFESSIONAL EXPERIENCE

Formal Verification Engineer Intern

Formal Land, Paris, France / Remote

Jun 2026–Present

[rocq-of-rust](#)[Public contributions](#)

- Updated rocq-of-rust for a newer version of Revm (a Rust implementation of the Ethereum Virtual Machine), adapting its Rust-to-Rocq translations, functional specifications, and proofs to changes in Revm’s architecture, and added evaluation through OCaml extraction.
- Currently developing instruction-level correspondence proofs between Revm and go-ethereum in Rocq, relating their models of stack, memory, storage, gas, host state, control flow, and errors. Also built mutation tests and runners for official Ethereum test cases.

Student Researcher

System Analysis and Verification Lab, NUST

Jun 2025–Present

[Playground](#)[Algorithms](#)

- Explored Lean 4 through game-based exercises and small formalizations, then moved to Rocq and its functional programming language, Gallina, for larger verification projects.
- Proved correctness properties of BST insertion, insertion sort, merging sorted lists, and MPI block-decomposition formulas, and formalized red-black-tree invariants and repair operations.
- **Currently working on:** Formal Verification of μC /OS-III kernel in Rocq and Verified Software Toolchain (VST).

Summer Research Intern

Deep Learning Lab, NCAI, NUST

Jun–Aug 2025

[Repository](#)[Certificate](#)

- Trained models across decision trees, neural networks, ResNet50, and YOLO on CIFAR and custom datasets
- Reproduced active-learning research for medical-image analysis and compared selection methods using TF-IDF and Sentence-BERT representations.

ACCEPTED RESEARCH

- [1] **Arooj Fatima and Saad Saghir.** “Towards Machine-Checked Verification of the μC /OS-III Kernel Using Rocq and the Verified Software Toolchain.” 2026

Verified with VST four C priority-bitmap functions for a pinned 32-priority μC /OS-III configuration. Generated their CompCert Clight representation using `clightgen` and proved initialization, insertion, removal, and highest-ready-priority lookup correct against abstract ready-priority specifications.

Accepted for presentation at the **FMCAD 2026 Student Forum.**

[Repository](#)

- [2] **Arooj Fatima, Saad Saghir and Adnan Rashid.** “A Comparative Formal Verification Framework for General-Purpose CPU Scheduling in Rocq.” 2026

Developed Rocq models and proofs for six CPU schedulers, including red-black-tree-based CFS and identified reusable proof patterns for minimum keys, layered priorities, and ticket intervals.

Accepted for publication at the **6th IEEE International Conference on Digital Futures and Transformative Technologies (ICoDT2 2026).**

[Repository](#)

OPEN-SOURCE CONTRIBUTIONS

Contributor to coqutil and Parseque	<i>Jul 2026–Present</i>
coqutil and Parseque	Rocq
• Authored seven merged pull requests adding build documentation, reusable characterization lemmas for lists, options, results, and length-indexed tuples, and multiline parser-position tests. • Contributions: #168, #169, #170, #171, #176, #178, and #22.	

PROJECTS & SELF-LEARNING

Software Foundations	<i>Dec 2025–Present</i>
<i>Rocq / Self-learning</i>	Private per authors' request
Working through selected exercises from <i>Logical Foundations</i> and <i>Programming Language Foundations</i> , covering inductive types, recursion, pattern matching, structural induction, higher-order functions, inductive relations, big-step operational semantics, program equivalence, semantics-preserving optimizations, nondeterminism, and Hoare logic.	
Verified Distributed Protein-Network Analysis	<i>May 2026</i>
<i>PySpark, Rocq, and STRING / Part of Parallel & Distributed Computing Course</i>	Repo
Processed 13.7M protein interactions into 473,860 high-confidence edges using PySpark; formalized and verified Large-Star and Small-Star at a fixpoint in Rocq.	
Biological Sequence Formalization	<i>Jan 2026</i>
<i>Rocq / Self-learning</i>	Repo
Formalized DNA/RNA sequence operations and proved properties of complementation, pairing, concatenation, repetition, reversal, and GC-content preservation.	
Concurrent Web-Server Simulation on FreeRTOS	<i>Dec 2025</i>
<i>C, FreeRTOS, RISC-V, and QEMU / Part of Operating Systems Course</i>	Repo
Built a concurrent web-server simulation demonstrating priority-based scheduling, producer-consumer task queues, and mutex-based synchronization for HTTP requests.	
Parallel Algorithm Performance Analysis	<i>Nov 2025</i>
<i>C, OpenMP, Pthreads, and MPI / Part of Operating Systems Course</i>	Repo
Implemented and benchmarked parallel matrix multiplication and merge sort on datasets ranging from 10 to 10 million elements, comparing the overhead and scaling behavior of three parallel-programming models.	
Operating Systems Concepts OSTEP	<i>Nov 2025</i>
<i>C / Self-learning</i>	Repo
Implemented OSTEP exercises on process API (fork/exec/wait), CPU scheduling algorithms, and memory virtualization.	
Standard ML and HOL4	<i>Jul 2025</i>
<i>Standard ML (SML), HOL4 / Self-learning</i>	Repo
Studied Standard ML and HOL4 through exercises involving recursion, currying, HOL terms and types, algebraic datatypes, and termination-checked function definitions.	
Movie Recommendation System	<i>December 2024</i>
<i>Python, Streamlit / Part of Object Oriented Programming Course</i>	Slides
Developed movie recommendation web app using Python and Streamlit with OOP principles.	

OTHER ACTIVITIES

Curriculum Design and Mentorship Team	<i>Oct 2025–Present</i>
<i>SPARK</i>	
Technical Writing and Research Resource Curation	<i>Jan 2025–Mar 2026</i>
<i>Published technical notes on TCS/FV/QC and curated research opportunities.</i>	Notion Medium Repo
Event Organization and Administration	<i>Oct 2024–Apr 2025</i>
<i>NUST Entrepreneurs Club / ICoDT2 2024 Conference</i>	Certificate Certificate

AWARD

FMCAD 2026 Student Forum Travel Award, Selected recipient	<i>2026</i>
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TECHNICAL SKILLS

Verification and Functional Programming: Rocq, Gallina, Lean 4, OCaml, Verified Software Toolchain (VST), CompCert/Clight, SML, HOL4	
Programming, Systems and Tools: Rust, C, Python, Git, Linux CLI, FreeRTOS, MPI, OpenMP, L ^A T _E X	